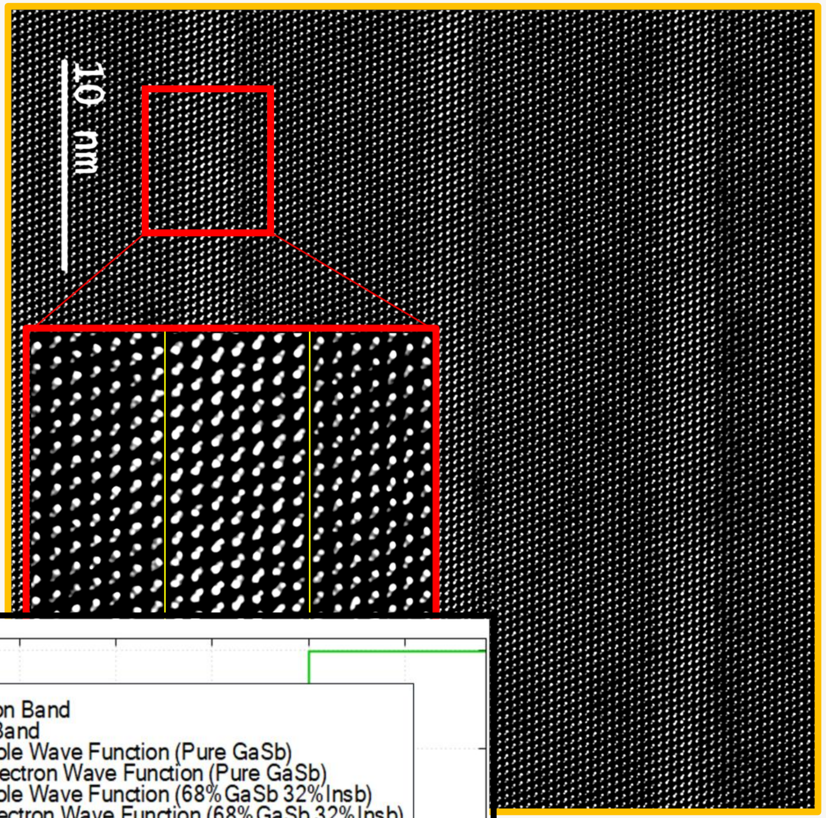
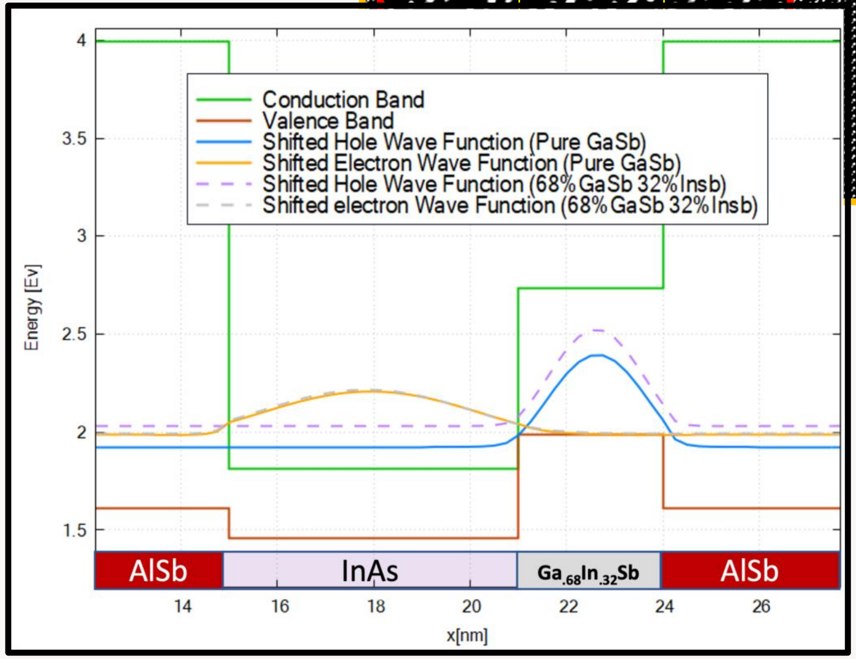
Epitaxial Engineering (111)-Oriented Antimonide Materials for Topological Quantum Systems

Dissertation Proposal

Jimmy Rushing
Department of Electrical and Computer Engineering
Tufts University





Scientific Motivation

The dissertation applies a shared workflow of MBE synthesis, ex situ structural characterization, and electronic modeling to the InAs/GaInSb and Sb/GaSb materials systems.

Antimonides combine **narrow gaps**, strong **spin-orbit coupling**, and **flexible** heterostructure design.

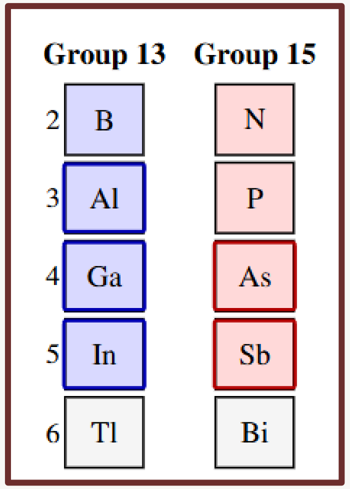
- Narrow-gap antimonides support infrared and broken-gap device physics
- Strong spin-orbit coupling makes them relevant to topological phases
- III-V molecular beam epitaxy provides precise control of thickness, strain, and interfaces

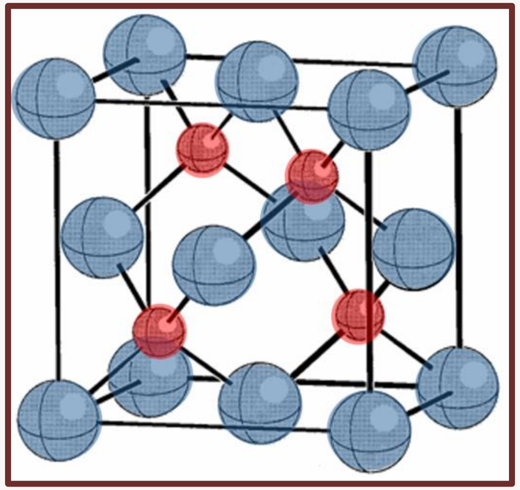
Why **(111)**?

- Distinct kinetic pathways can enable leverage over interface formation and mixed-anion transitions
- Distinct electronic symmetry and band structure
- Offers new leverage over interface formation and mixed-anion transitions
- May open a more favorable route to Sb heteroepitaxy
- Experimentally fertile and untilled ground

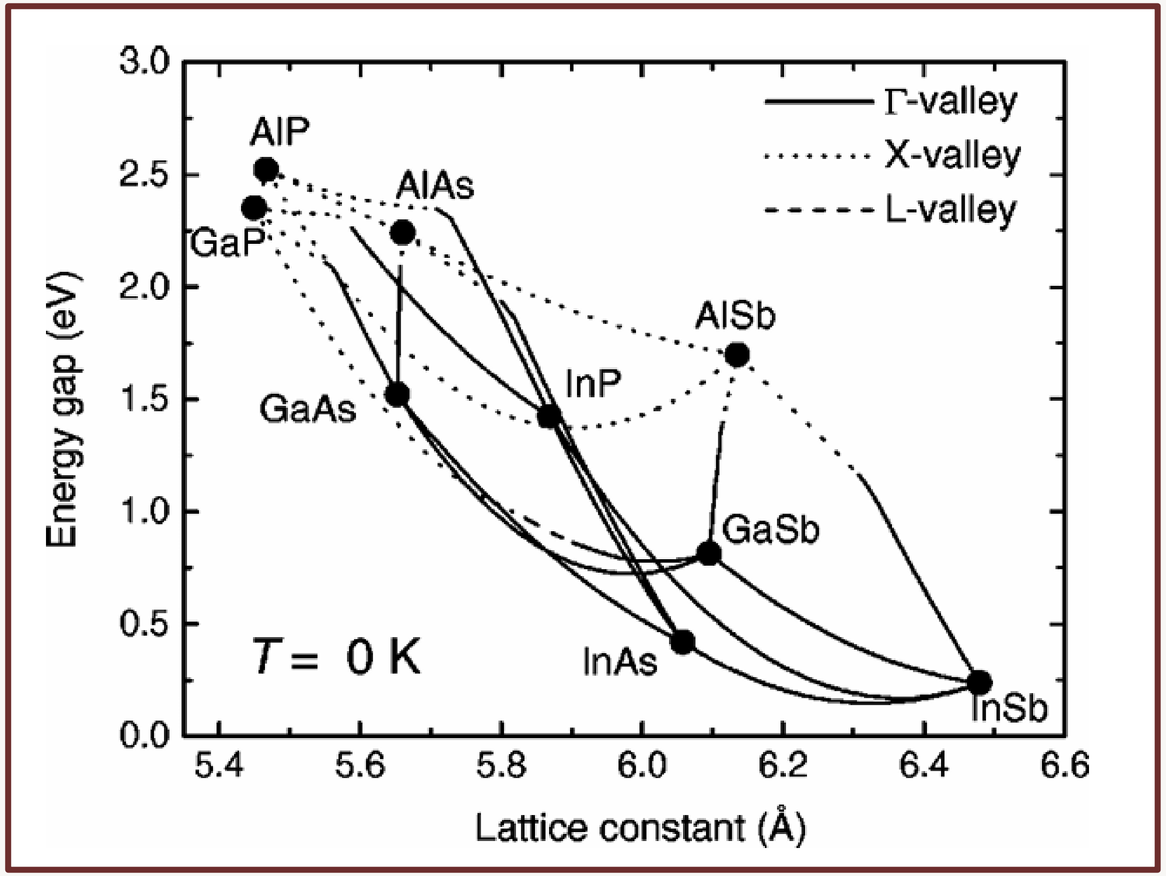
III-V materials

- Semiconductors
- Mature epitaxy and processing
- Strong spin-orbit coupling
- Band Gap engineering through compositional continuity
- Better device integration
- Facilitate heterogenous integration and QW growth to further engineer bands

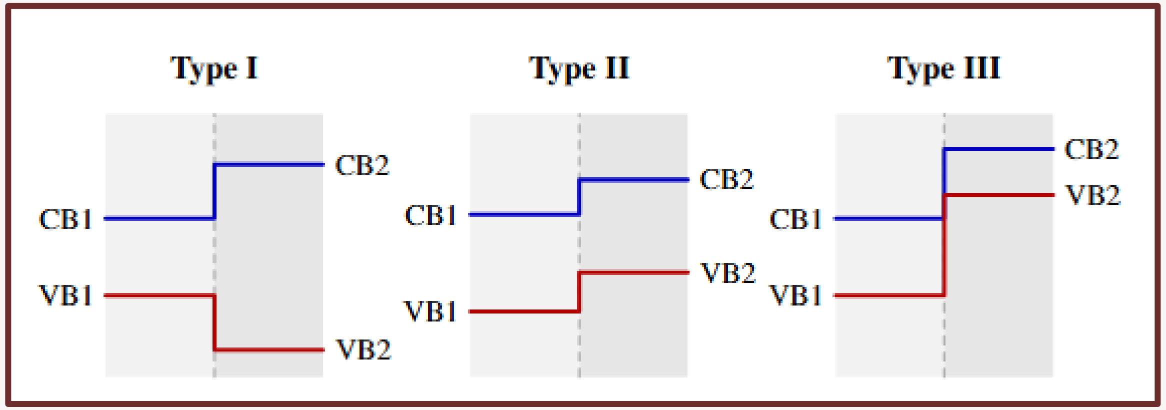




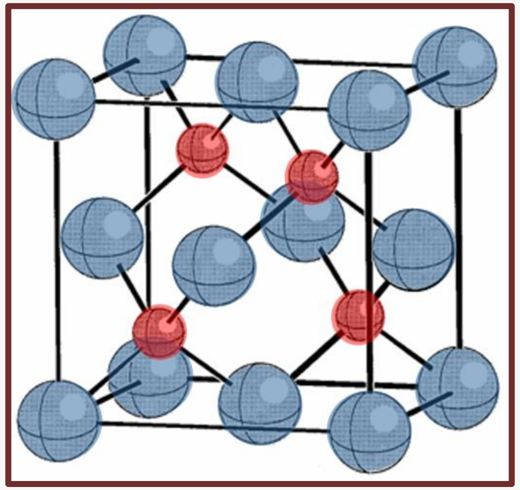
G. S. Rohrer, *Structure and Bonding in Crystalline Materials* (2001)



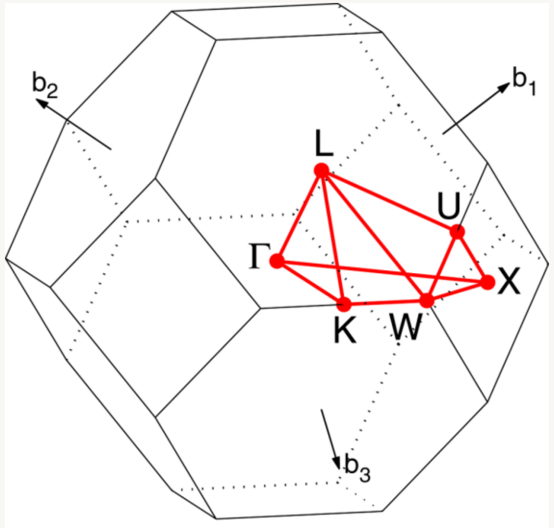
I. Vurgaftman, J. R. Meyer, and L. R. Ram-Mohan, *J. Appl. Phys.* 89, 5815 (2001)



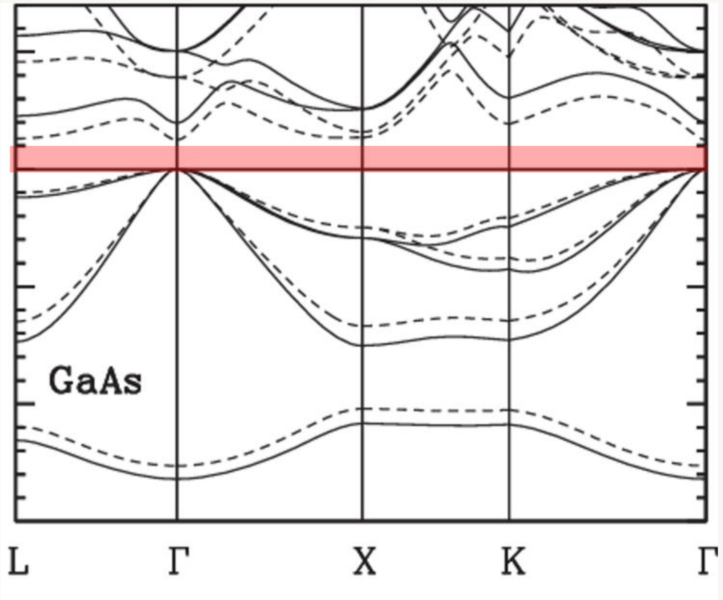
From Crystal Structure to Band Structure in Semiconductors



G. S. Rohrer, *Structure and Bonding in Crystalline Materials* (2001)



Wikimedia Commons, “Face-Centered Cubic Lattice (Brillouin zone)”



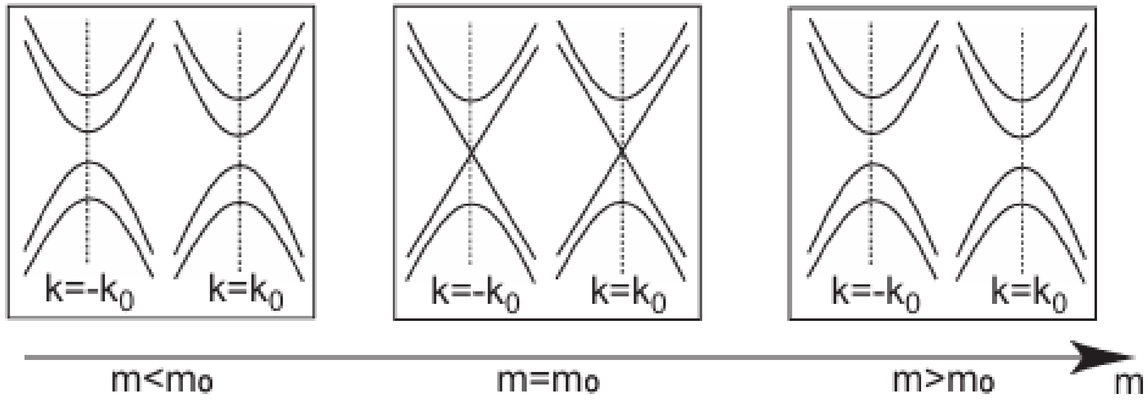
Campo and Cococcioni, *J. Phys.: Condens. Matter* 22, 055602 (2010)

- Crystal periodicity gives Bloch states
- Reciprocal-space symmetry defines the Brillouin zone
- The eigenvalues of the Bloch Hamiltonian form the bands

What is a Topological Insulator?

It’s a fancy semiconductor

- The occupied-band structure cannot be smoothly deformed into that of a trivial insulator
- At an interface to a trivial material, the topological class must change
- That change requires local gap closure, which produces conductive edge or surface states



Murakami, *Physica E* 43, 748-754 (2011)

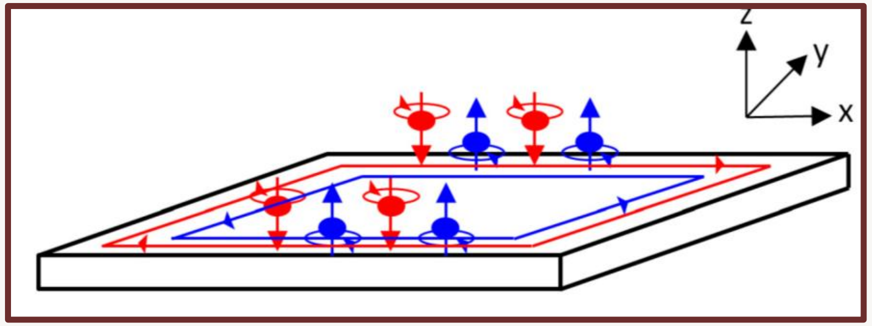
Topological insulators are special because a boundary to a trivial insulator **must** host conductive states.

What is a Quantum Spin Hall Insulator?

A 2D ($\partial = 2$) topological insulator with time-reversal symmetry (Class = AII)

Class	T	C	S	δ = 0	1	2	3	4	5	6	7
A	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0
AIII	0	0	1	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
AI	+	0	0	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$
BDI	+	+	1	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$
D	0	+	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0
DIII	−	+	1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$
AII	−	0	0	$2\mathbb{Z}$	0	$\boxed{\mathbb{Z}_2}$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0
CII	−	−	1	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0
C	0	−	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0
CI	+	−	1	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$

- 2D topological insulator results from spin orbit coupling
- Bulk insulating gap in the plane, edge conduction:
  the interior is gapped, but the sample edge remains conducting
- Helical edge states:
  opposite spins propagate in opposite directions along the same edge
- Time-reversal protection:
  nonmagnetic backscattering is suppressed because the edge modes form a Kramers pair



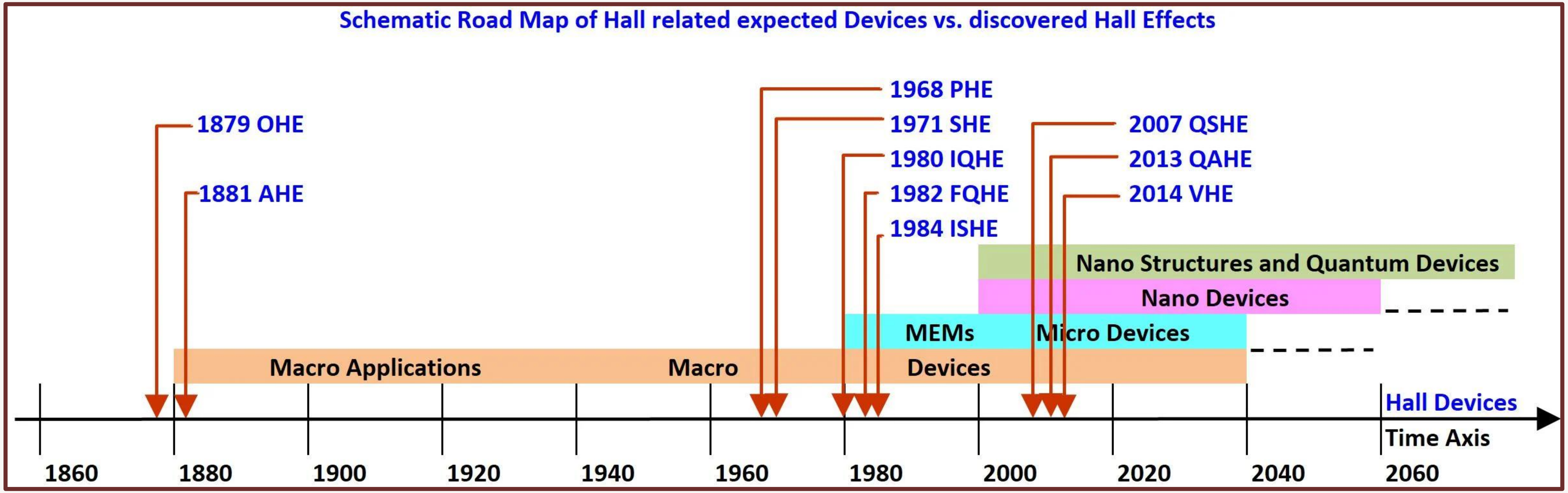
Karsenty, *Sensors* 20, 4163 (2020), doi:10.3390/s20154163

Device implications

- Low-dissipation edge transport
- Platform for topological devices (spintronics, quantum computing, interconnects, and sensing)

Hall-family roadmap to topological transport

This dissertation targets the quantum spin Hall branch through epitaxial InAs/GaInSb quantum wells and Sb-based thin films

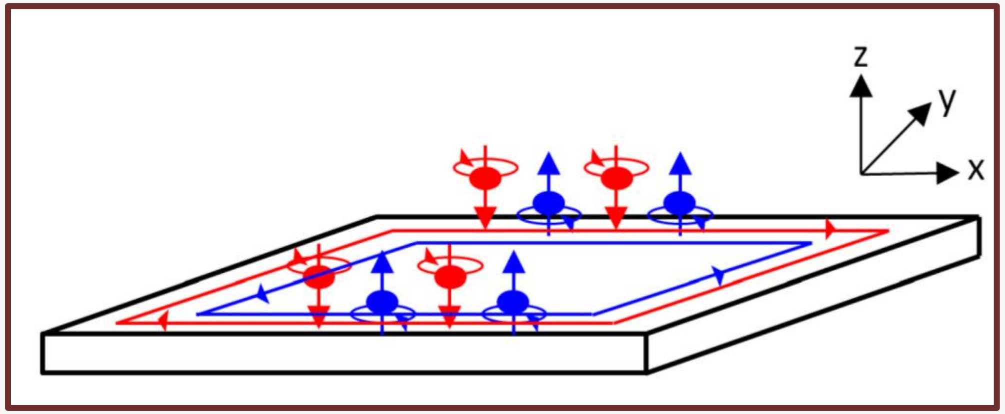


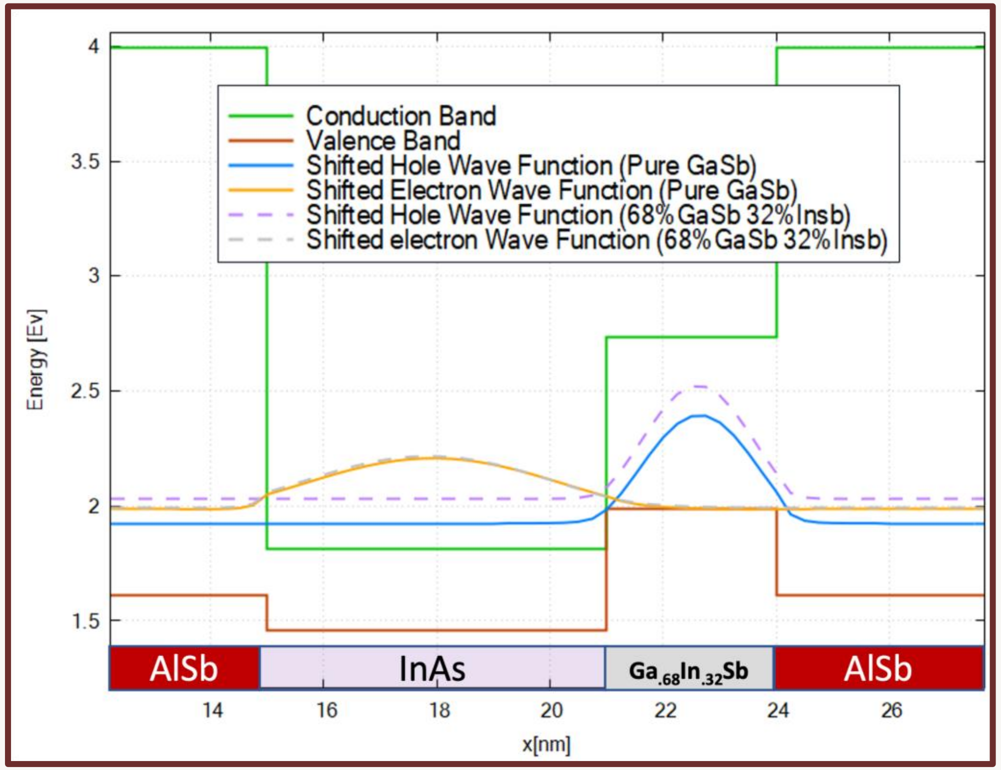
Karsenty, *Sensors* 20, 4163 (2020), doi:10.3390/s20154163

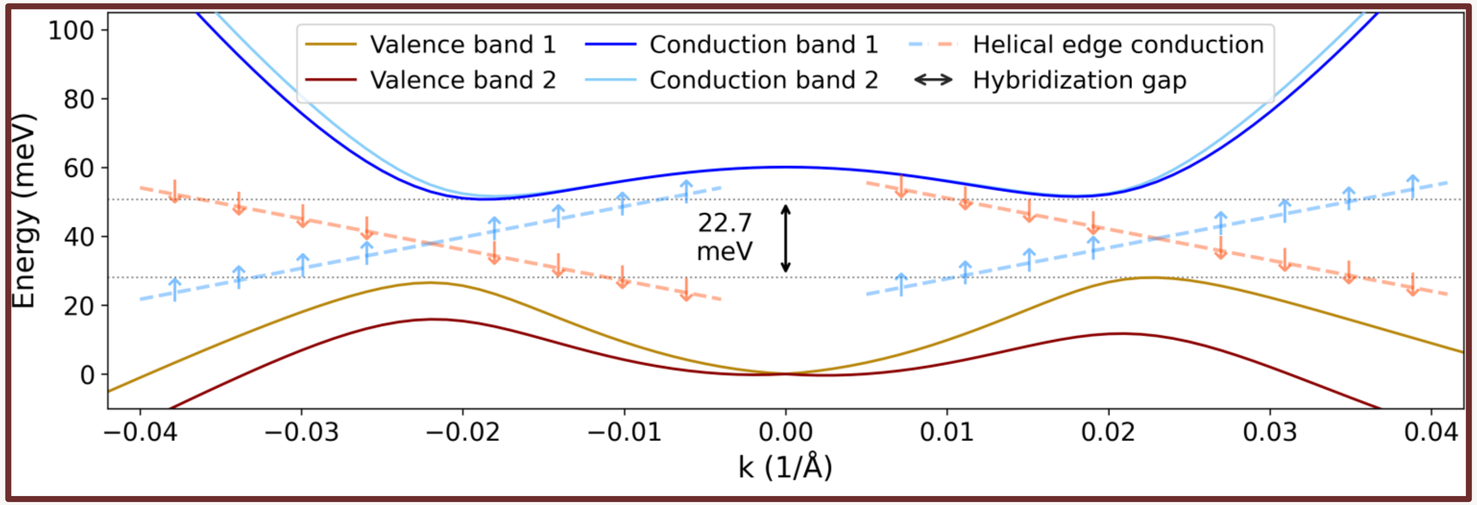
QSHI from III-V materials

Why? How?

- Tune subbands with thickness and strain
- Use alloying to shift band alignment
- Control interface chemistry by MBE
- Engineer hybridization with layer sequence
- Refine structure through epitaxial design
- Electrostatically tunable topological gaps
- Cleaner to work with than Hg-based systems



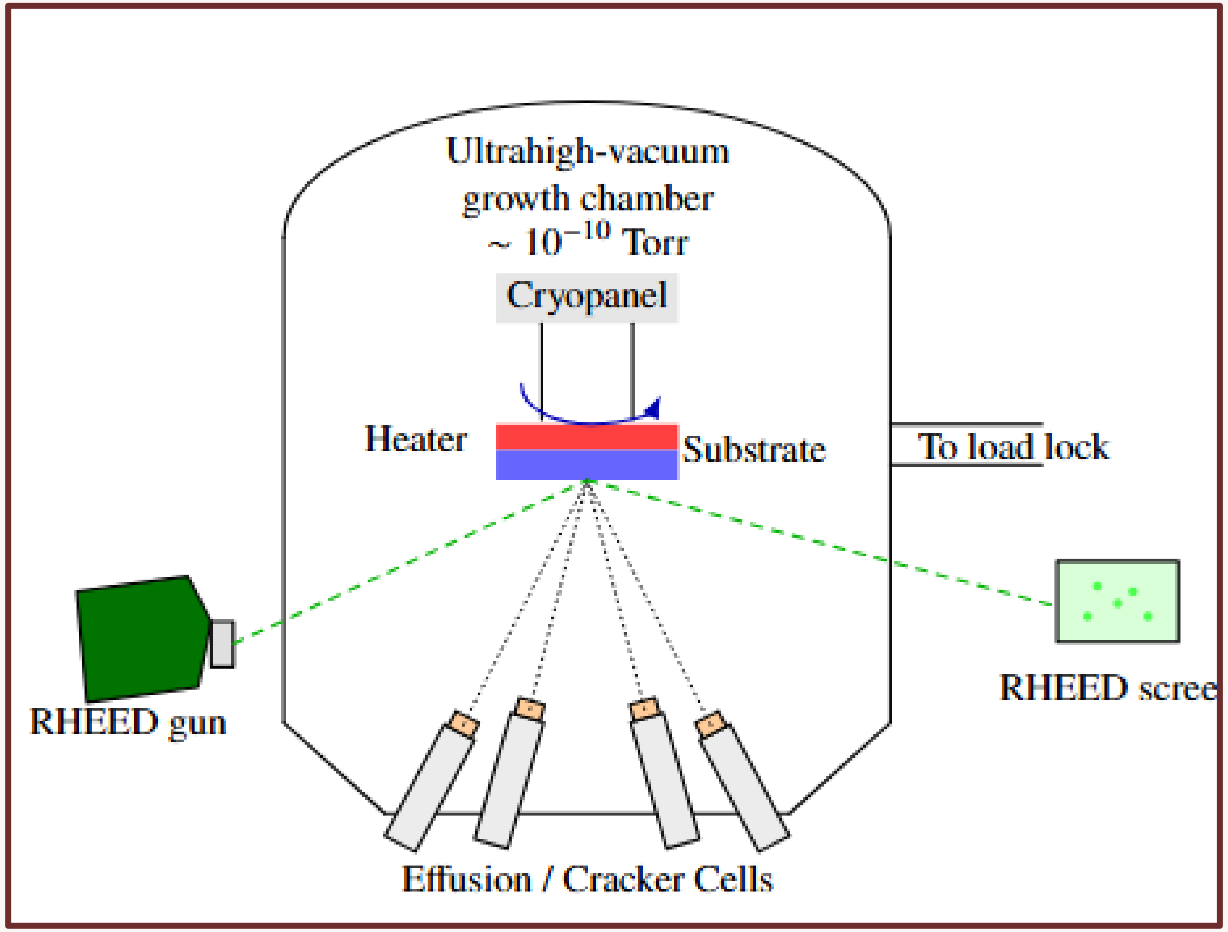




III-V Material Synthesis

Why Molecular Beam Epitaxy (MBE)?

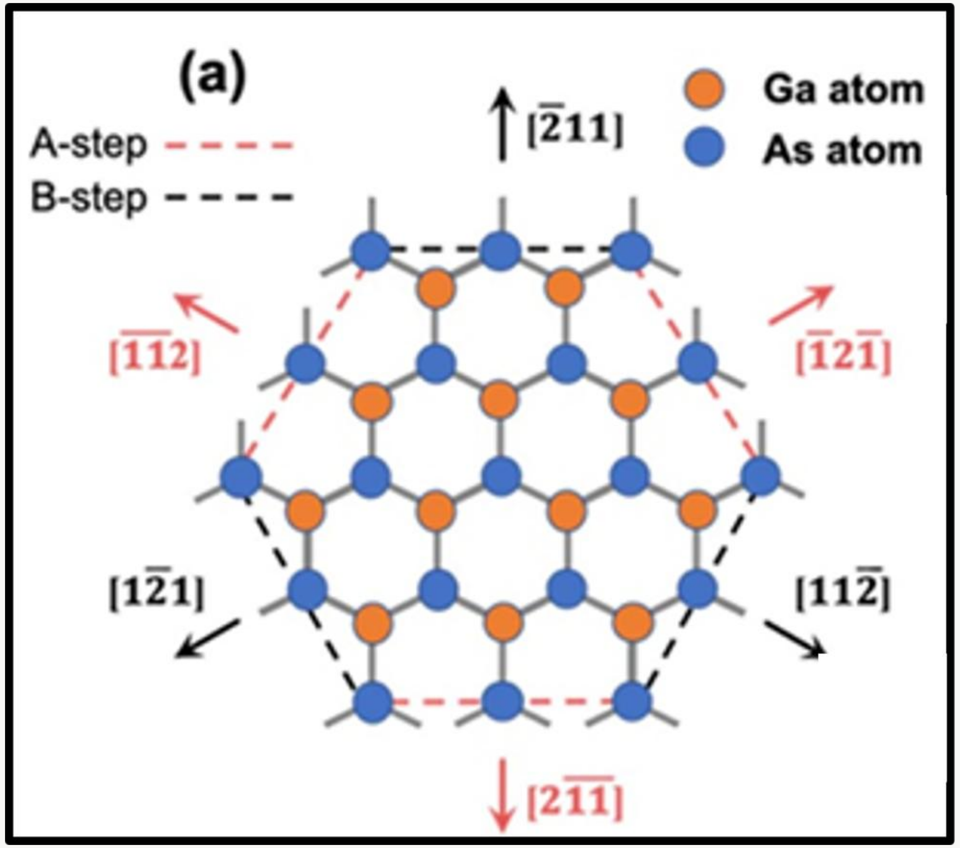
- Monolayer-scale thickness control
- Control of strain via composition
- Precise layer sequencing
- Direct control of interface chemistry
- Control of group V Species (e.g. As_4 or As_2)
- Well suited to mixed-anion antimonide systems



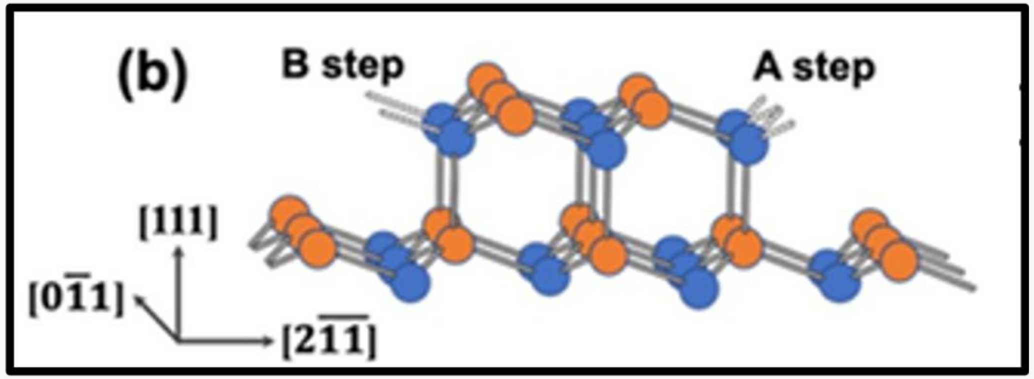
QSHI from III-V materials

Why pursue the 111-orientation?

- Distinct quantum-well symmetry from (001)
- Polar threefold surface symmetry
- Can suppress D’yakonov-Perel’ relaxation
- Potential for greater electrostatic tunability
- Inequivalent A- and B-type steps
- Anisotropic diffusion and roughening
- Group-V chemistry tunes step kinetics
- Morphology, interfaces, and electronic structure are tightly coupled



Schuck et al., *J. Vac. Sci. Technol. B* 38, 032801 (2020)



Orientation-Specific In Situ characterization and Process Steps

- (001) and (111) have different surface packing and symmetry
- GaSb(001) is a well-studied III-V MBE reference surface
- Dual mounting exposes both orientations to the same growth conditions
- This bridges established (001) process knowledge to (111)A
- It also enables direct comparison of reconstruction and morphology

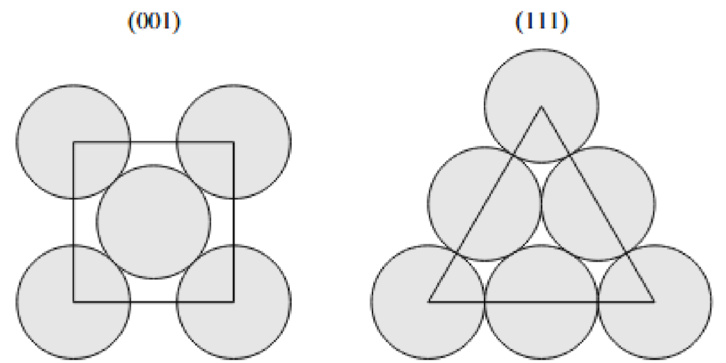
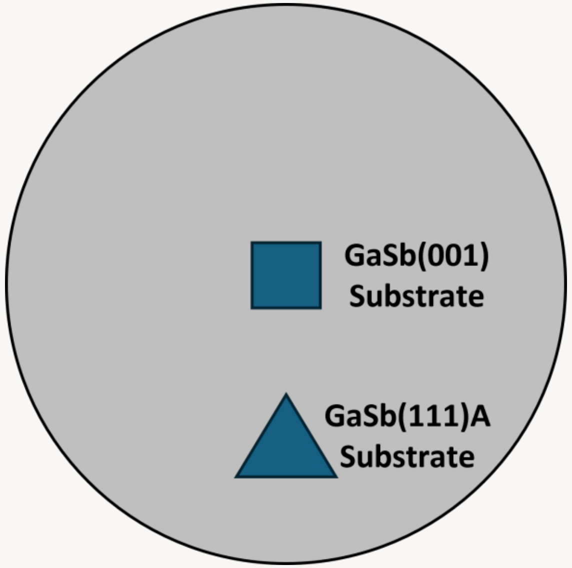
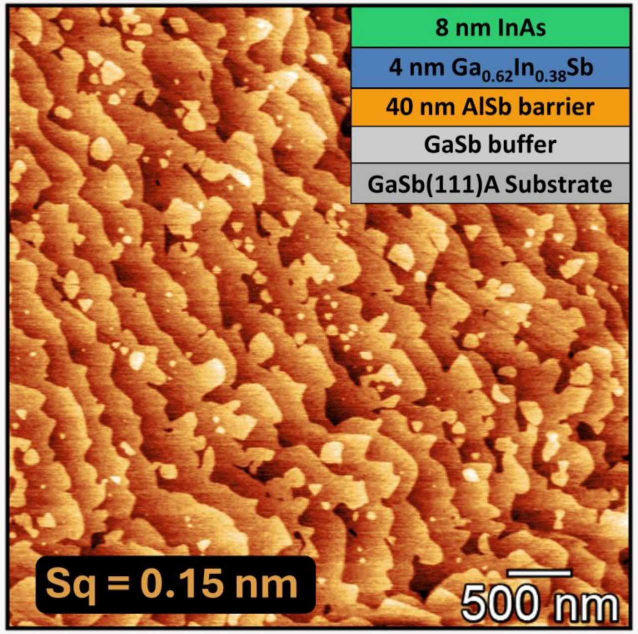


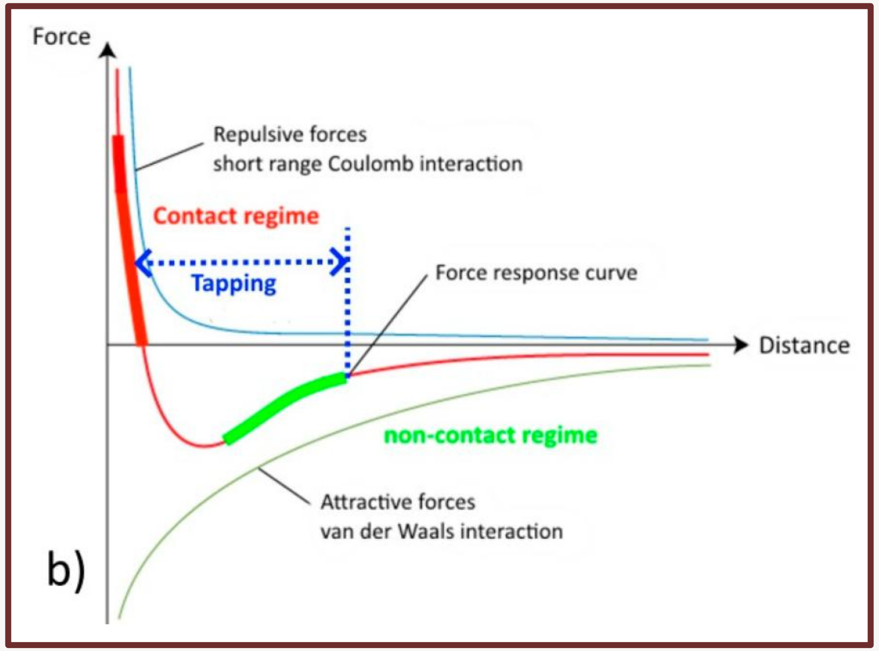
Figure 10: Schematic comparison of atomic packing in zincblende (001) and (111) planes.

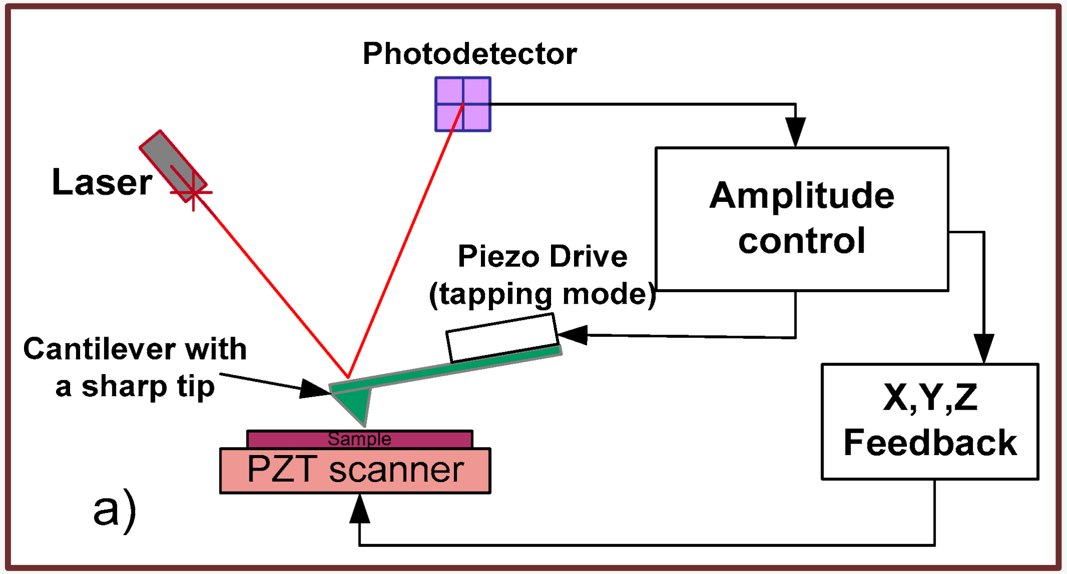


Atomic Force Microscopy

- Measures real-space surface morphology via cantilever force detection (optical lever readout)
- Tapping-mode operation minimizes lateral shear → preserves surfaces
- Enables identification of kinetic regimes: 3D islanding, terrace recovery, surfactant-mediated smoothing
- Quantifies morphology using areal RMS roughness $Sq = \sqrt{(1/A \int (z-\bar{z})^2 \, dA)}$



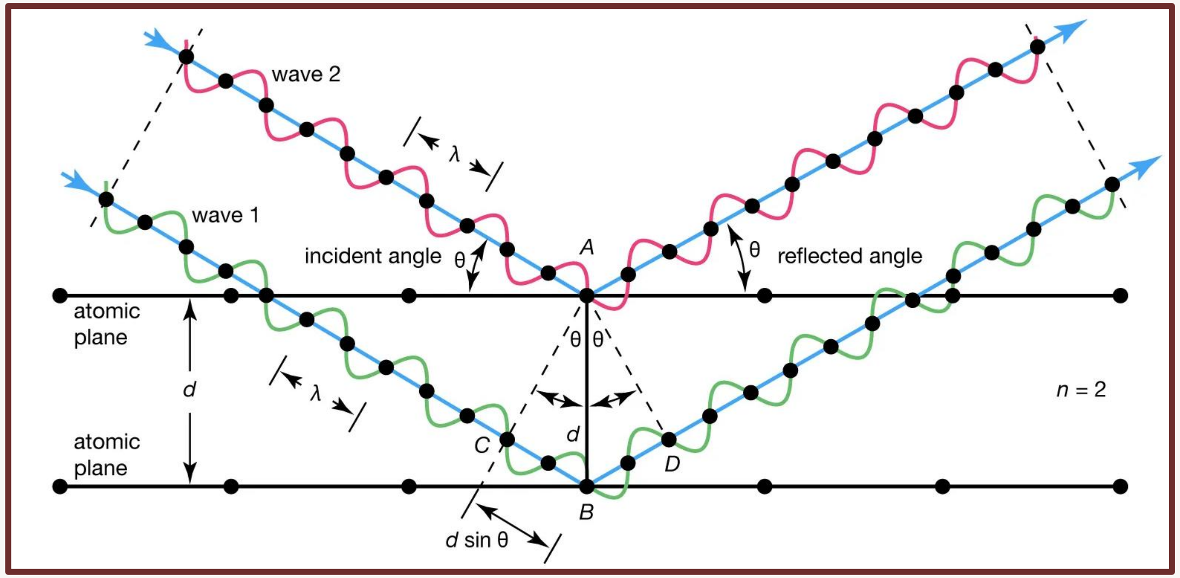


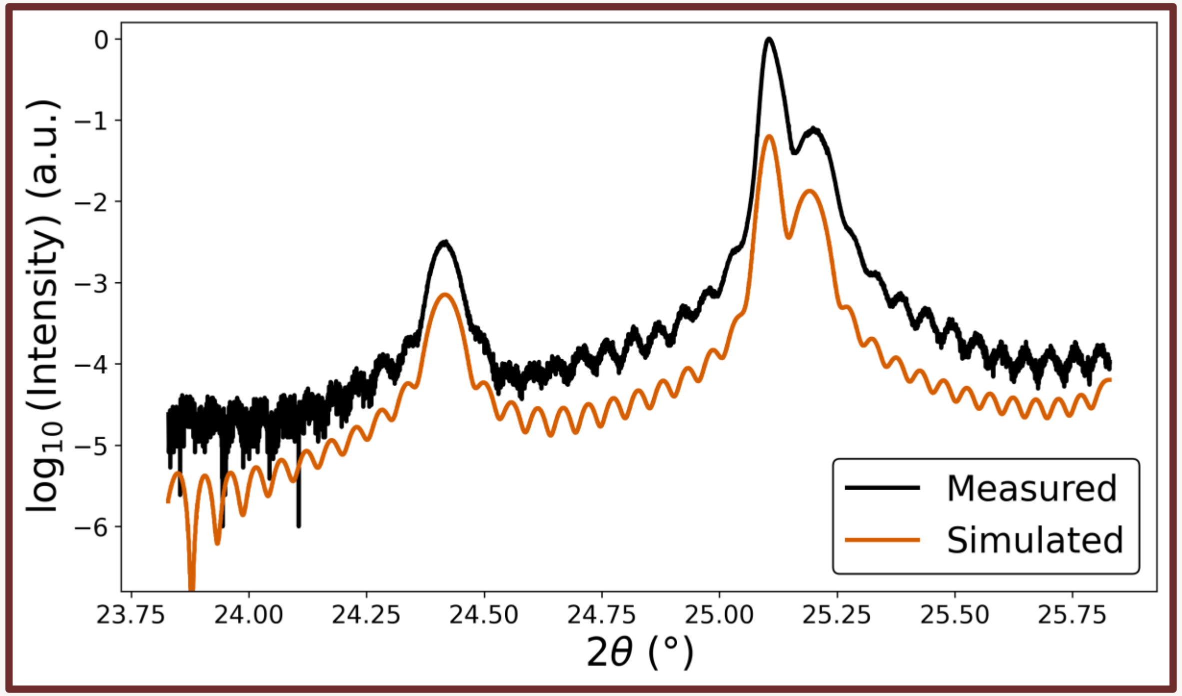


•K. Yasakau, *Corros. Mater. Degrad.* 1(3), 345-372 (2020).

High-Resolution X-ray Diffraction and Reflectivity

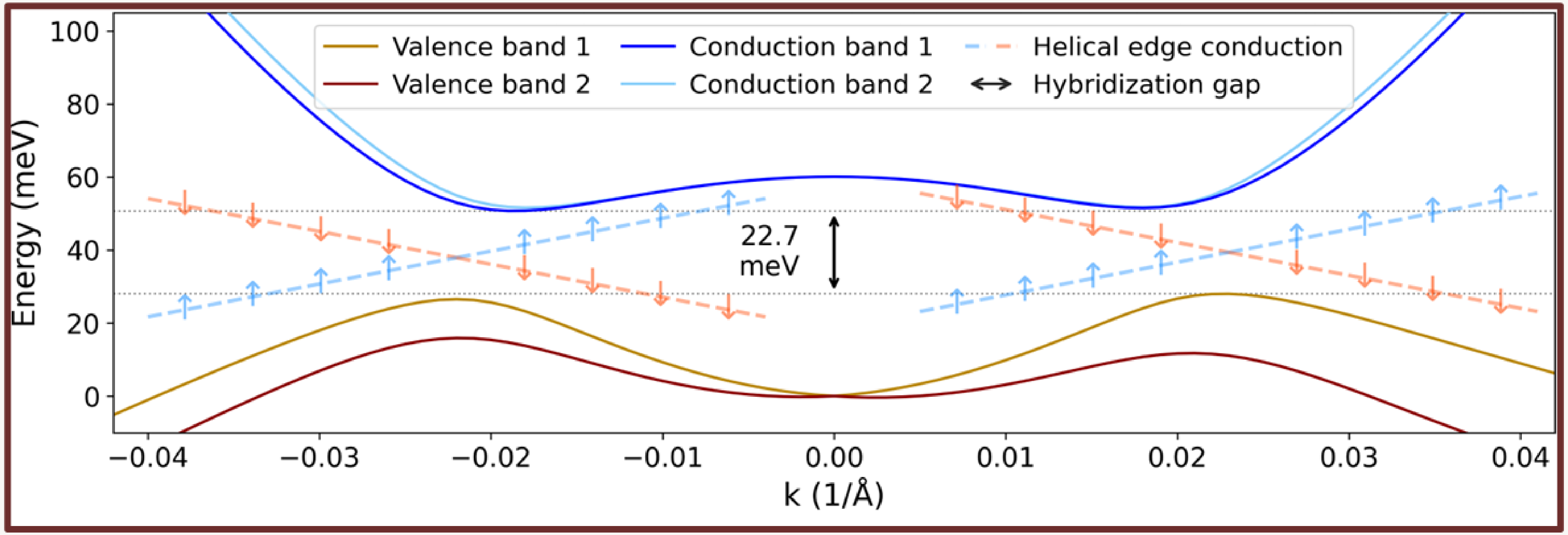
- Bragg's law: $2d \sin\theta = n\lambda$ links diffraction angle to lattice-plane spacing
- Symmetric (111), (222), (333), and (444) reflections constrain out-of-plane lattice parameter and strain state
- Satellite peak spacing gives superlattice bilayer period; envelope asymmetry reveals grading and thickness redistribution
- Pendellösung fringes arise from coherent interference
- XRR complements HRXRD with rapid thickness, density contrast, and roughness measurements





Simulation Toolbox

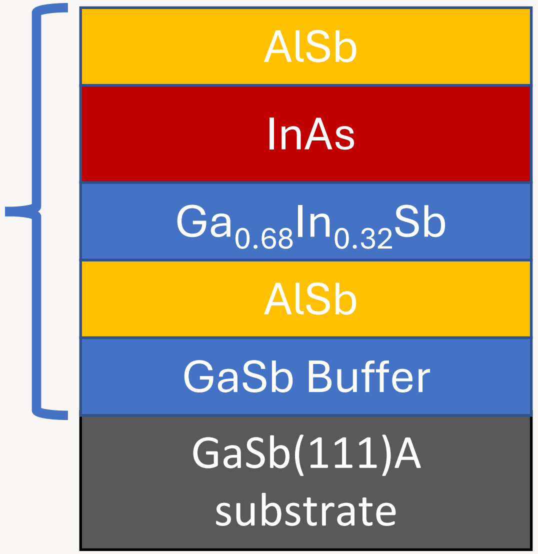
- Nextnano for band profiles and wavefunctions
- CADtronics for finite-k subband dispersion
- 8-band and 14-band k·p / Kane modeling
- Tests effects of thickness, strain, and interface abruptness

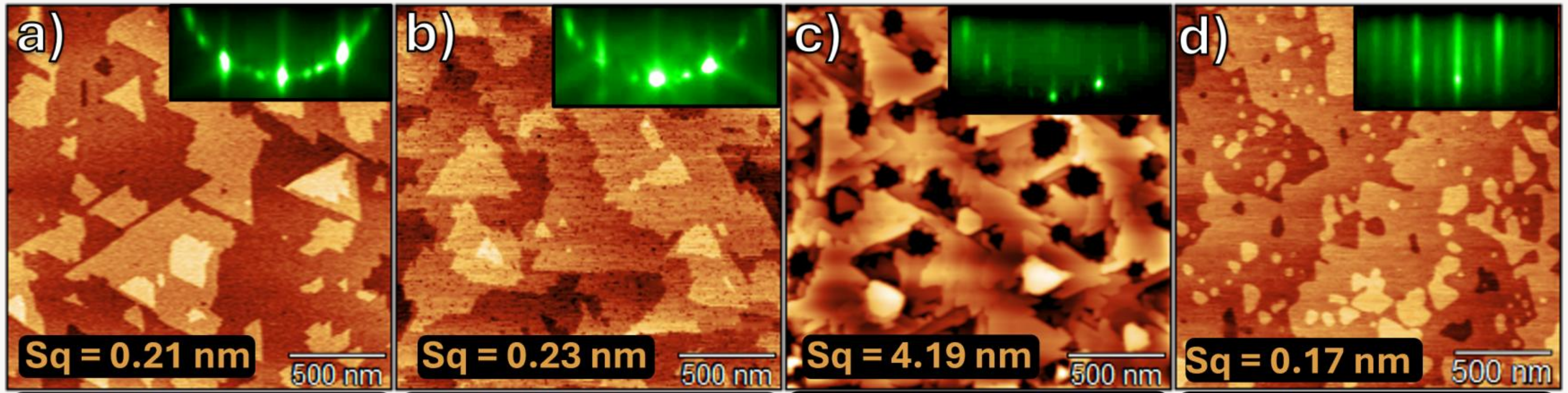


Constituent-Layer Optimization for InAs/ $Ga_{0.68}In_{0.32}Sb$ on GaSb(111)A

- Re-established GaSb(111)A growth on the Tufts MBE
- Optimized GaSb, $Ga_{0.68}In_{0.32}Sb$, and InAs constituent layers
- Established growth conditions for the active heterostructure constituents
- Identified $Ga_{0.68}In_{0.32}Sb$ as the morphology bottleneck

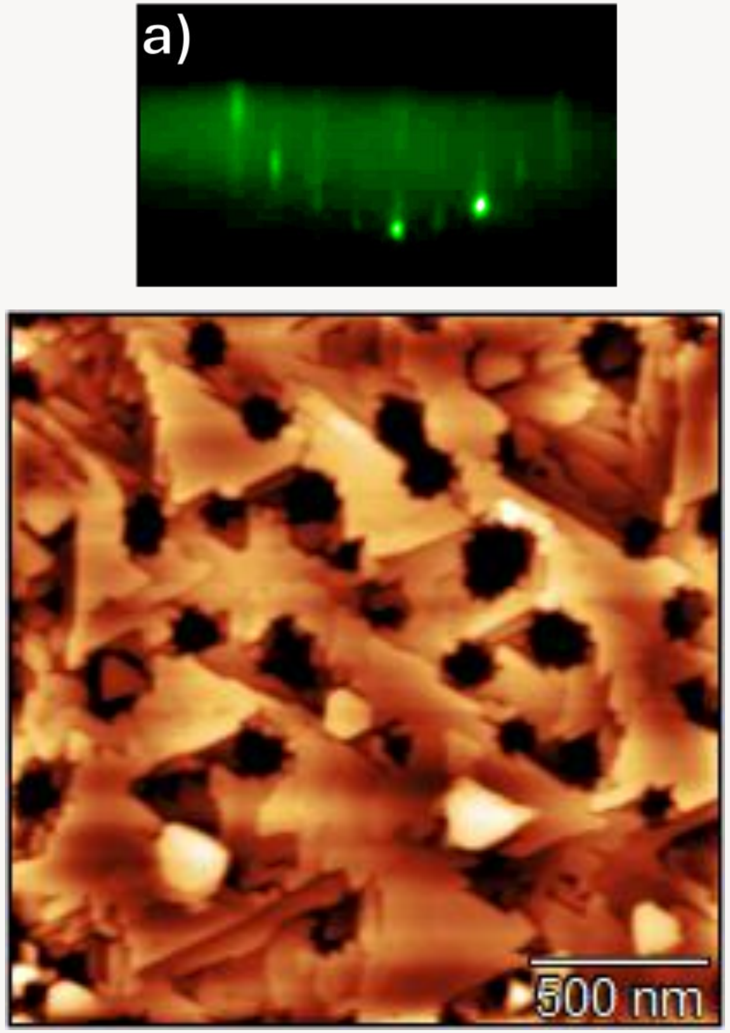
The ingredients

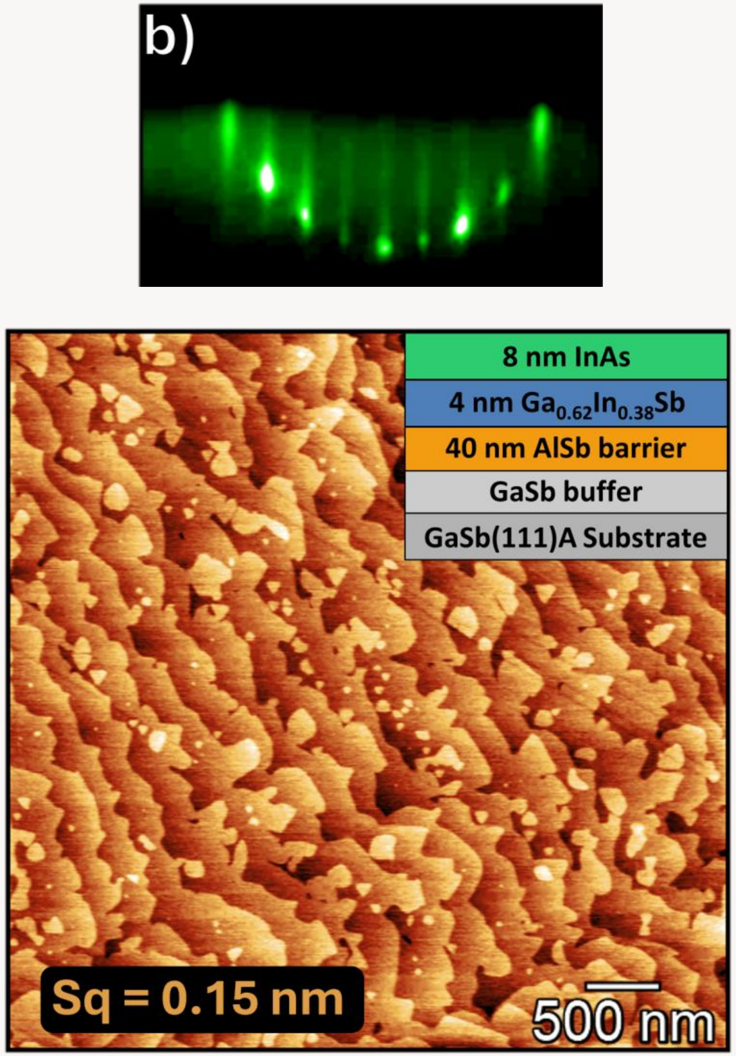




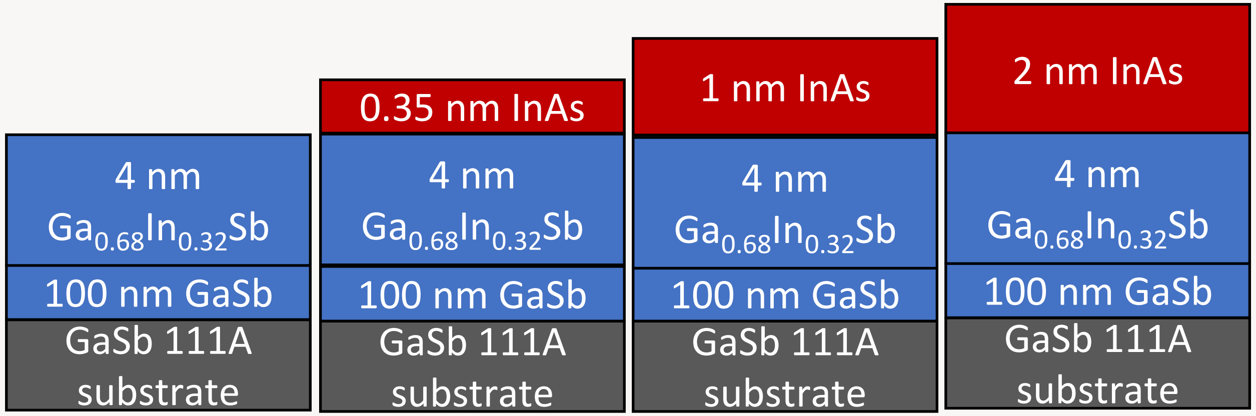
GaSb(111)A cooled under Sb_2 | GaSb(111)A cooled under As_4 | $Ga_{0.62}In_{0.38}Sb$(111)A cooled under Sb_2 | InAs(111)A cooled under As_4

Astounding Smoothing of $Ga_{0.68}In_{0.32}Sb$ Upon InAs Deposition

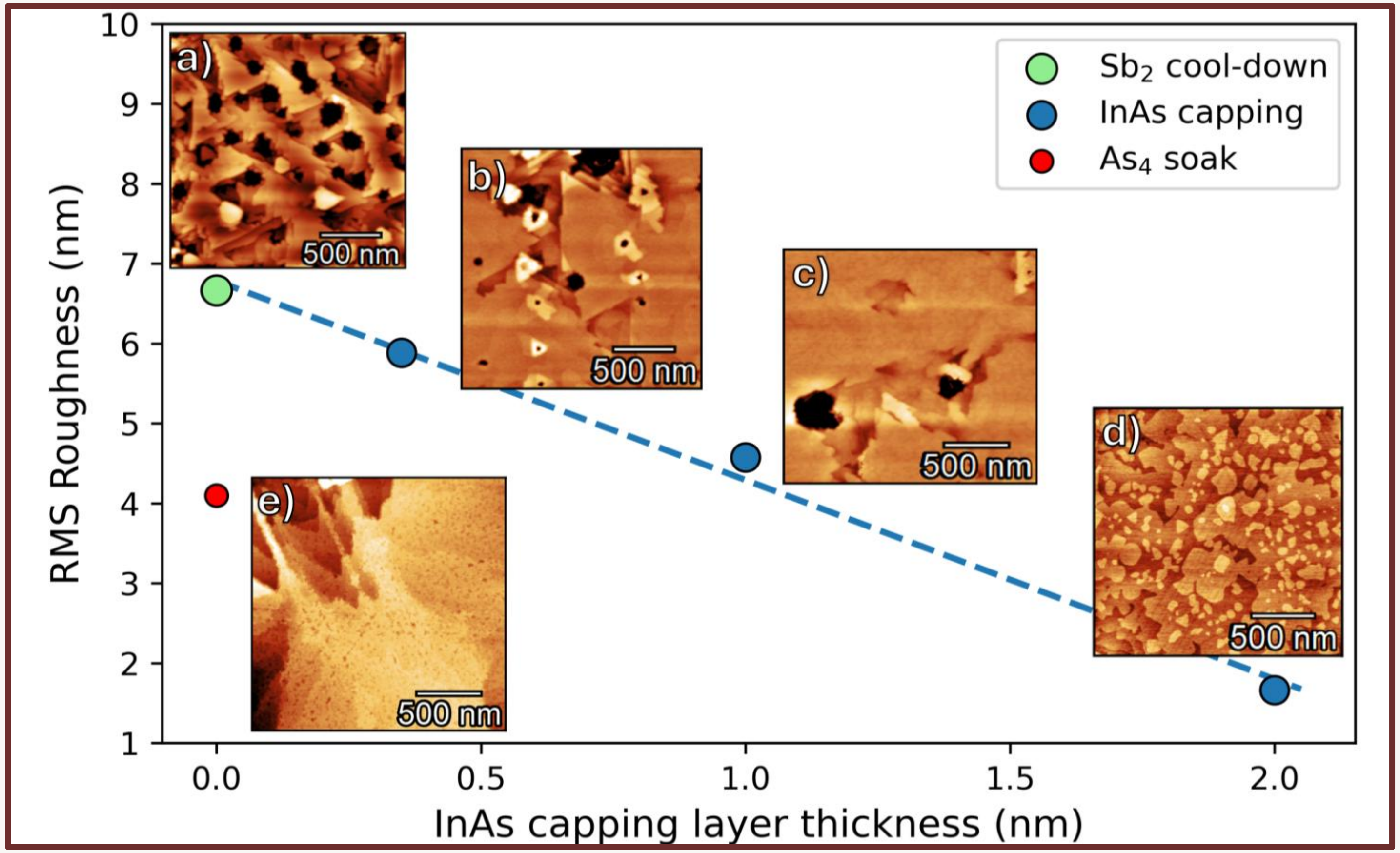




A Closer Look at the $Ga_{0.68}In_{0.32}Sb$ to InAs Transition

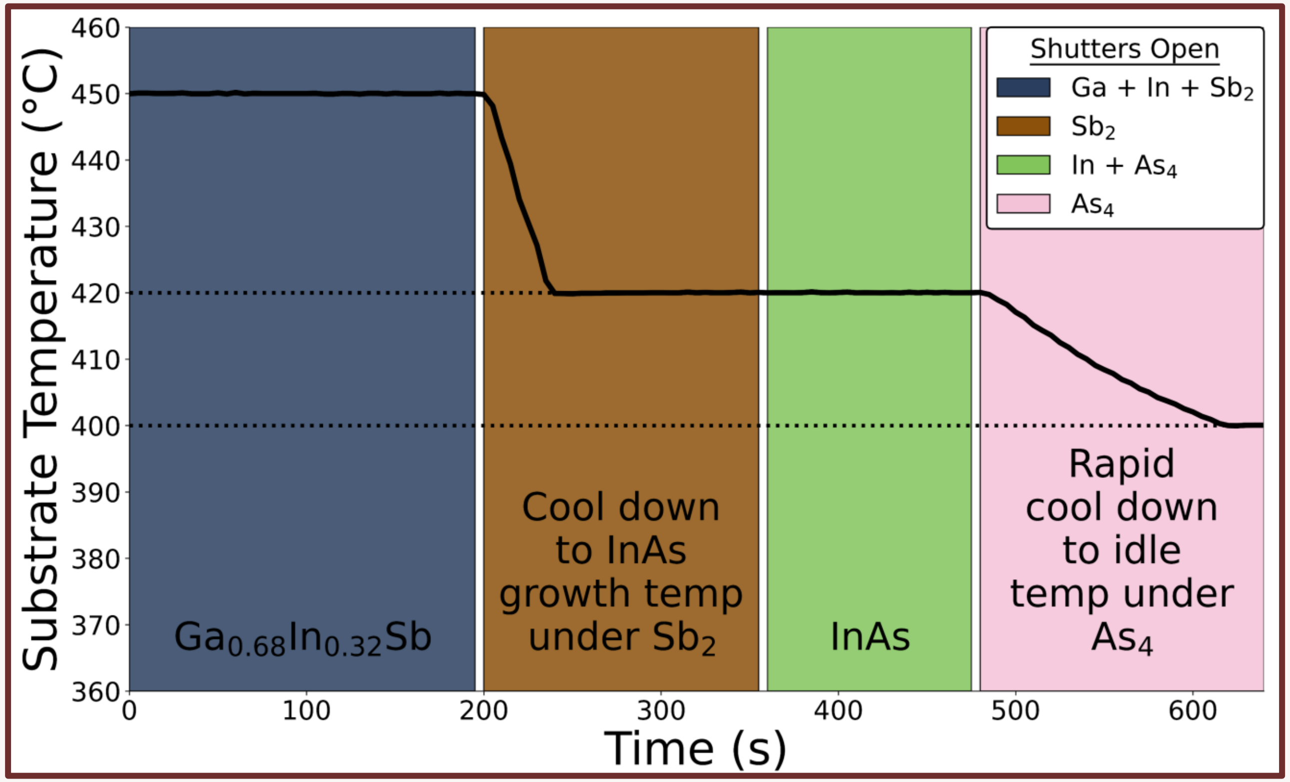


- Bare 4 nm $Ga_{0.68}In_{0.32}Sb$ remains strongly rough
- As4 soak alone reduces roughness, but only partially
- Progressive InAs capping produces monotonic smoothing
- 2 nm InAs yields the strongest morphology recovery
- The $Ga_{0.68}In_{0.32}Sb$ -to-InAs transition is a key kinetic control point



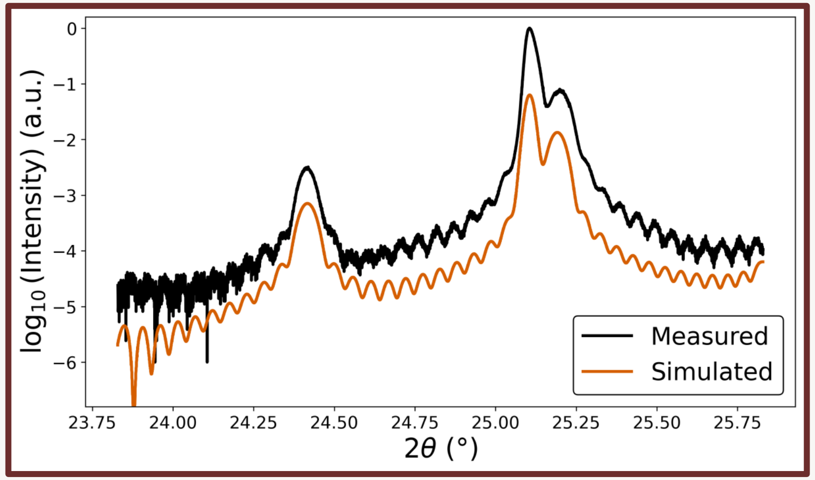
Growth Sequence for the Transition Experiment

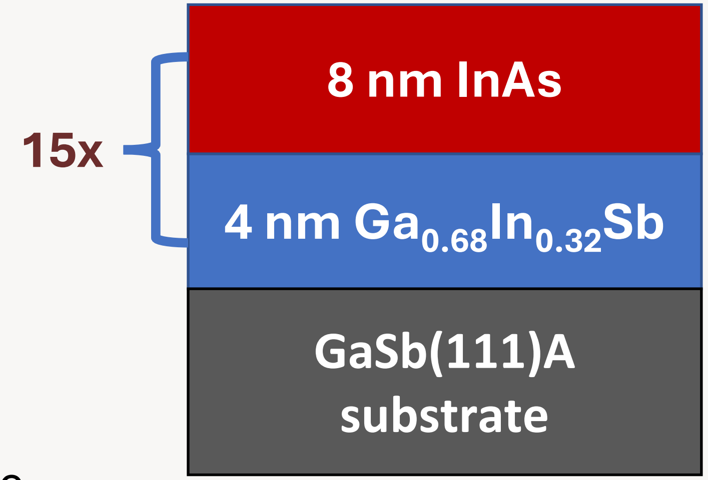
- 4 nm $Ga_{68}In_{32}Sb$ grown at 450 °C
- Cooled under Sb2 to the InAs growth temperature
- InAs deposited at 420 °C
- Rapid cooldown to idle under As4

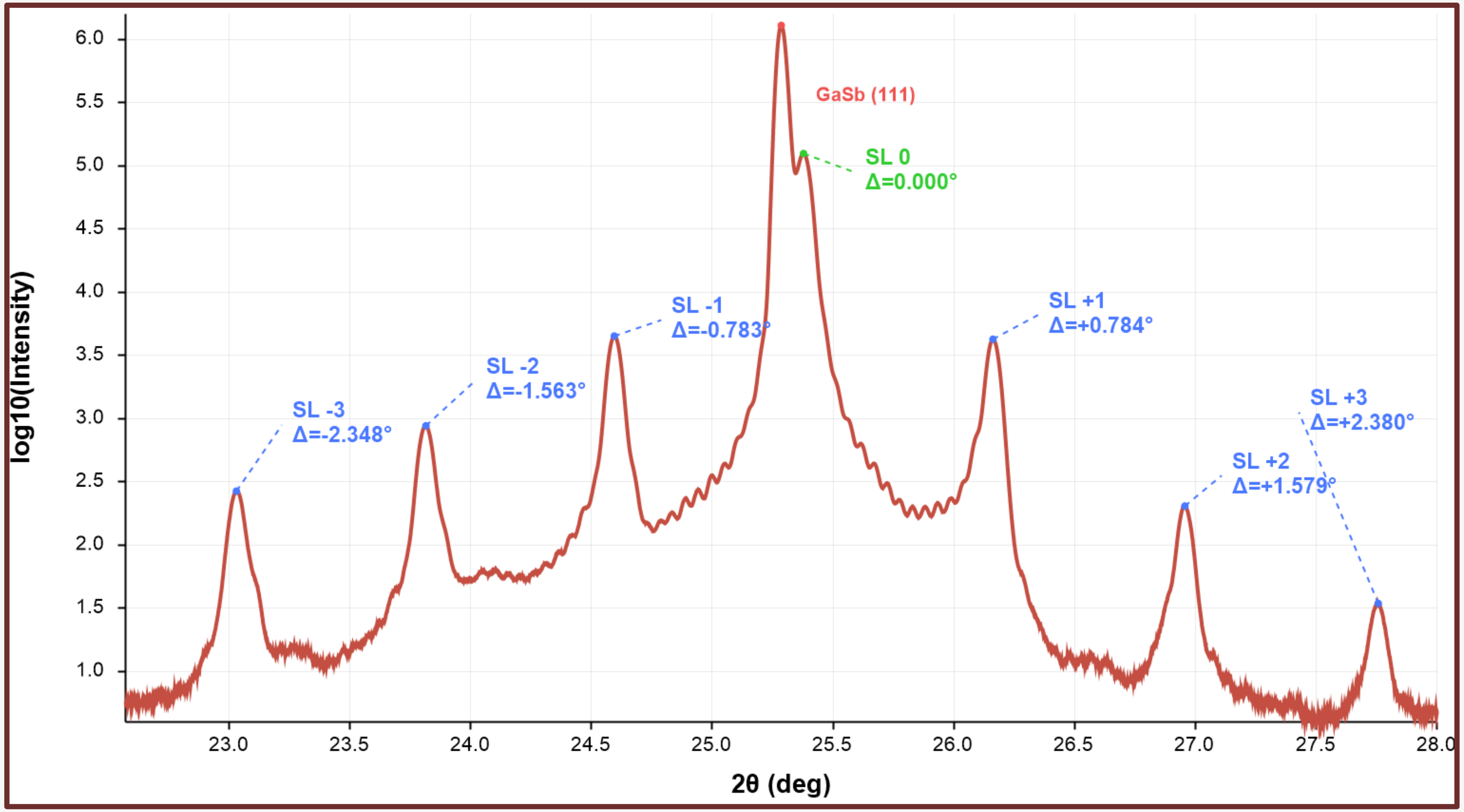


Structural Characterization of InAs/ $Ga_{0.68}In_{0.32}Sb$ Superlattices

- Symmetric satellite peaks confirm periodic growth
- Pendellösung fringes indicate coherence
- Simulation requires a thin graded interface
- Coherent, but not perfectly abrupt

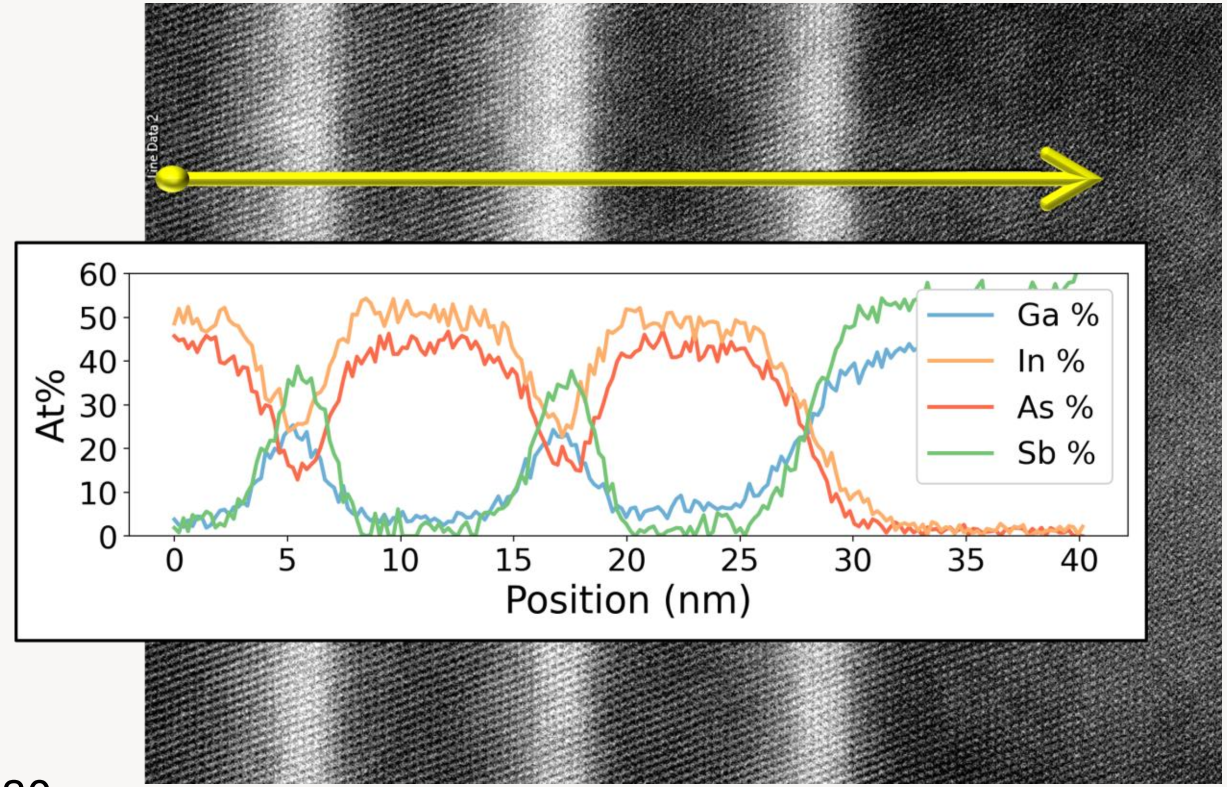


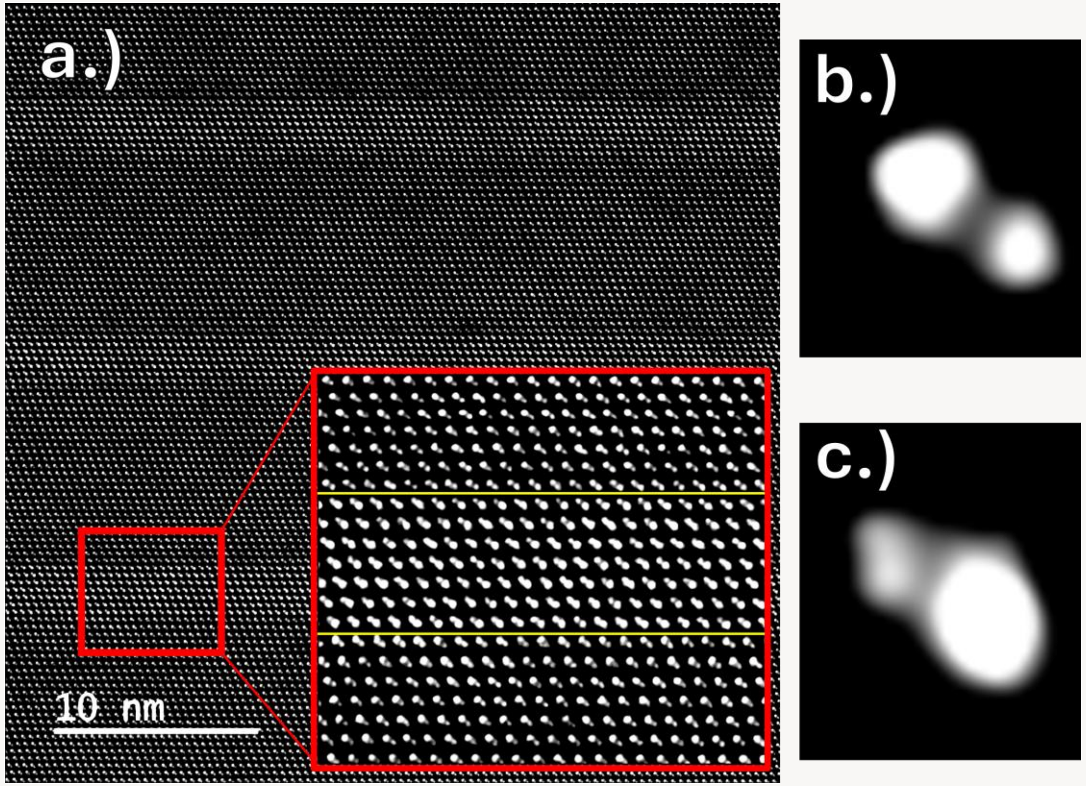




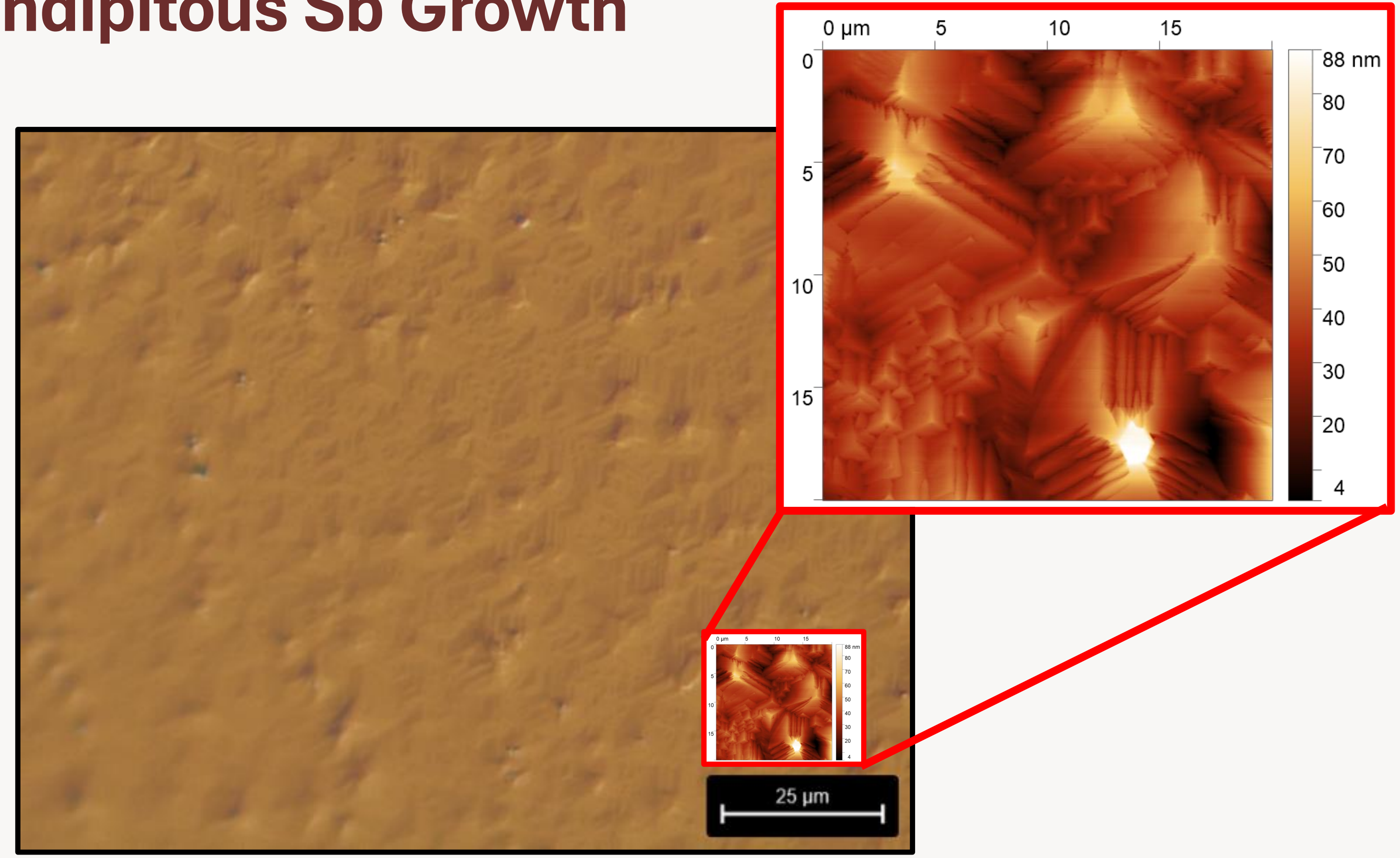
TEM and EDS Reveal Abrupt Interfaces with Slight Interfacial Asymmetry

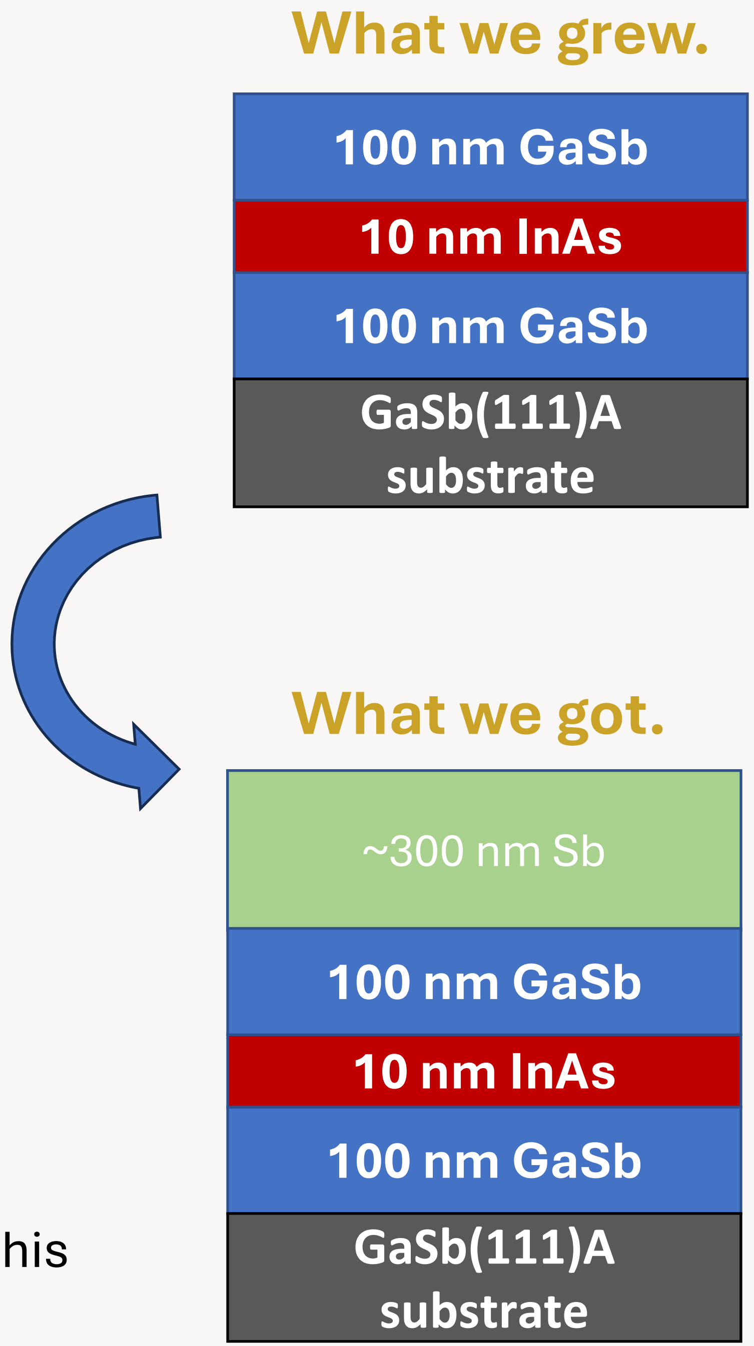
- TEM and EDS resolve distinct InAs and $Ga_{0.68}In_{0.32}Sb$ layers
- InAs-to-GaInSb interfaces appear sharper
- GaInSb-to-InAs interfaces appear more diffuse
- Interface asymmetry matches the AFM smoothing picture
- Coherent stack, but kinetically inequivalent interfaces





A Serendipitous Sb Growth

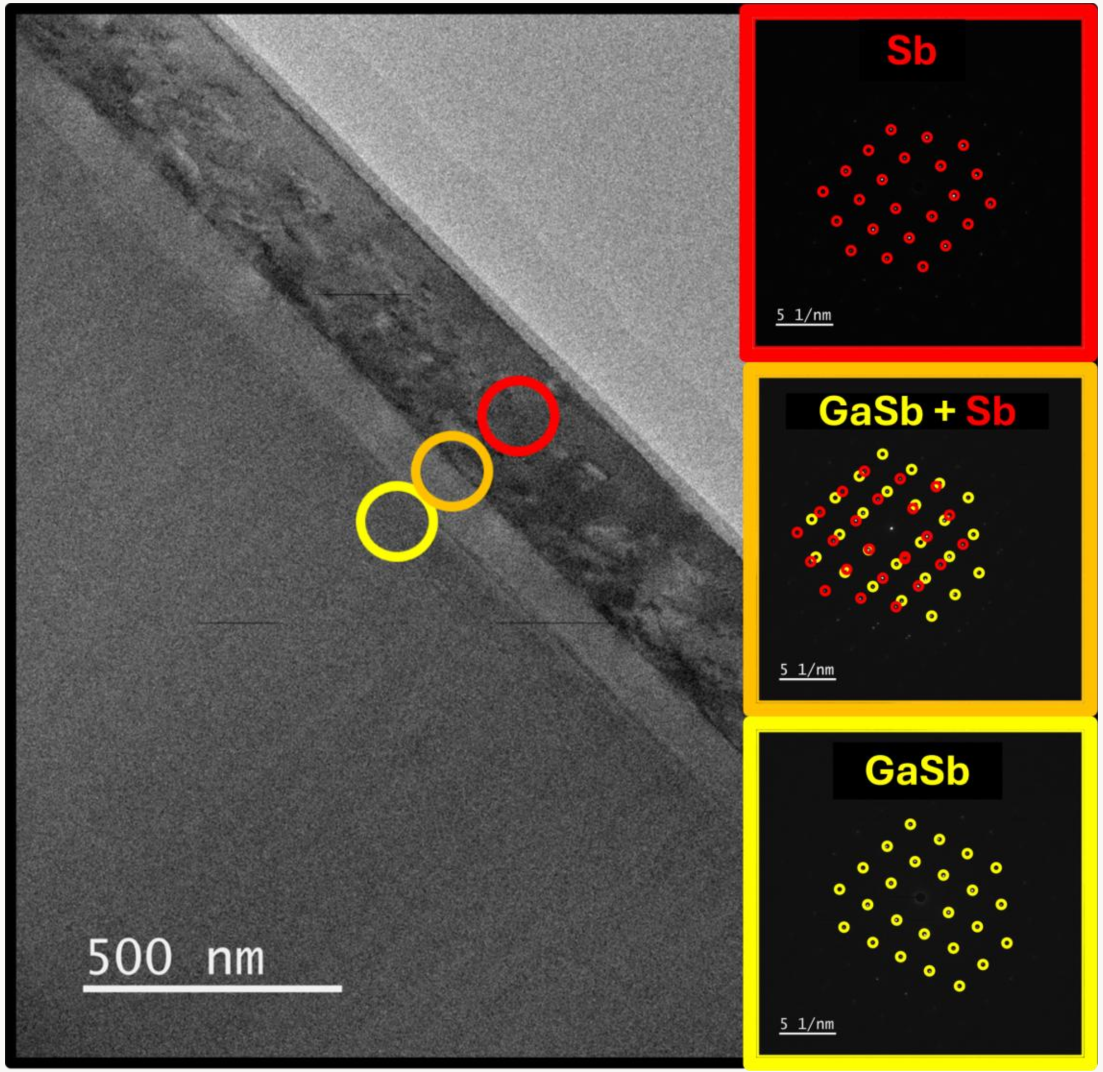


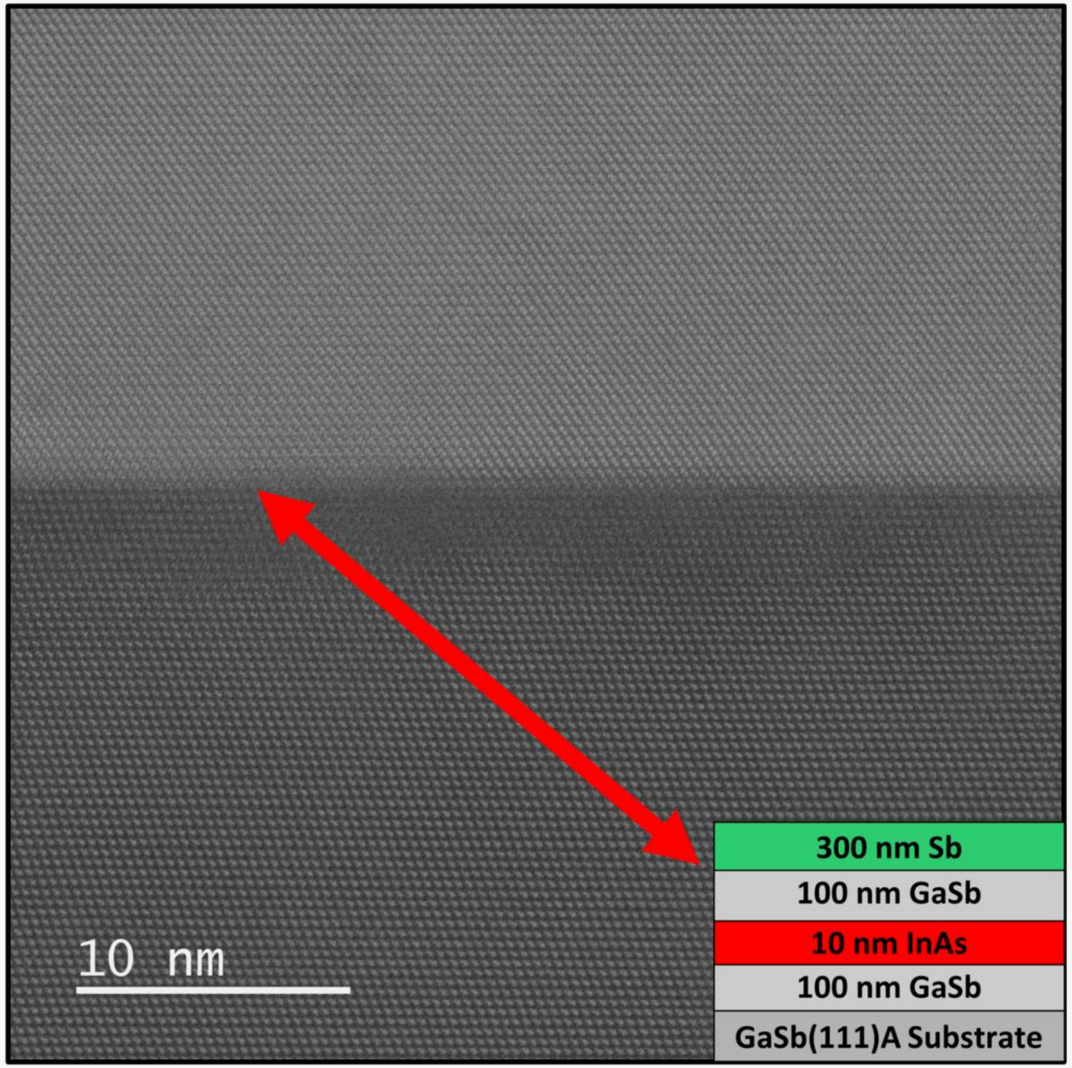


- Nominal III-V optimization sometimes produced rough surfaces
- No obvious in situ sign of catastrophic growth failure
- Post-growth AFM reveal rough but crystalline surface
- TEM analysis reveals source of roughness and motivates the rest of this dissertation

Elemental Sb: First Structural Evidence

Selected-area electron diffraction identifies rhombohedral Sb on GaSb(111)A, while high-resolution TEM shows a coherent interface with preserved crystallographic registry.





Quantum-Well Platform: Remaining Interpretation Work

Structural realism now needs to be carried through transport and modeling

What remains

- Complete magnetotransport interpretation with the University of Würzburg collaboration
- Use TEM/XRD-constrained thicknesses and interfaces in Kane-model calculations
- Benchmark 8-band nextnano against 14-band Cadtronics
- Test whether interface asymmetry degrades the intended hybridization gap

Key output

Relate measured structure to subband ordering, spin splitting, and finite-k hybridization

Determine whether realistic trilayers retain their intended symmetry in practice

Why this matters

The dissertation is not only about coherent growth, but about the realized electronic structure of those coherent stacks.

Mechanism-Guided Sb(111) Growth Plan

The favorable high-temperature onset is treated as the clue to how ultrathin Sb should be controlled

Central logic

- Final morphology alone does not identify a unique microscopic mechanism
- Nucleation-onset temperature is the key mechanistic diagnostic
- The first goal is to identify which variable shifts that onset most strongly
- That same variable is likely the one that most directly controls ultrathin-film quality

Candidate causes of the higher-temperature onset

- Terminating surface chemistry
- Sb2 versus Sb4
- Vacuum / process improvements
- RHEED beam effects during nucleation

Goal

Explain why the present Sb/GaSb system behaves so differently from the older low-temperature literature.

Sb Experimental Progression

Mechanism-discriminating comparisons first, optimization second

1. Starting surface

Compare direct, As-prepared, and InAs-prepared GaSb(111)A templates

Ask whether nucleation onset shifts as strongly as thicker-film morphology

2. Species / beam tests

Take the best-onset template as baseline

Run matched Sb2 vs Sb4 and RHEED on/off ultrathin growths

3. Window mapping

Map continuity, roughness, and retained thickness through the temperature range

Use AFM, XRR, and ellipsometry as the main decision metrics

4. Stability + calibration

Test thermal survival on return toward GaSb growth temperatures

Then complete flux-dependent growth-rate calibration

Stretch goals

- Test whether optimized Sb can support renewed GaSb overgrowth
- Explore whether Bi incorporation can be introduced into the Sb layer
- Work with Cadtronics to add Elemental Sb to modeling capabilities
- Work with Würzburg to find first observed TI signature in thin film Sb